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**One-Sample Tests (Z & T) Transcript**

Hello everyone, welcome to lesson 2 video 3. In this video, we will apply hypothesis testing framework to one sample test. These tests allow us to check if sample mean is consistent with the hypothesised population mean. The main topics that we will be studying in this video are Z-test and T-test.

In our last session, we built our foundation on hypothesis testing framework. Today, we are moving from theory to practise. We are going to apply the framework to one of the most statistical tasks that we have.

That is, testing a claim about a single population mean. This session is all about choosing the right tool for the job. We will learn the differences between two workhorse statistical tests.

One is Z-test and the other one is T-test. And most importantly, we will learn exactly when to use which. Let's get started.

So, last time, in the last video, you learned about applying the entire hypothesis framework as a 4-step process. In the very first step, you state the hypothesis. Remember, we had stated what is alternate, what is null and everything like.

Then you set the significance. You define the significance level, alpha, which generally is 5% or 10%, whatever you would want to have. Then you calculate the test statistics.

Then you make a decision. So, it's a 4-step process. Now, what we are going to do is, we are going to focus on step 3 for a specific scenario.

We have collected the data and that data basically is a one sample and we want to test a claim about the mean of the population using that sample. The core question we will answer in this is, which specific test should we use to get to the test statistic? The choice between the Z-test and T-test comes down to one simple piece of information about our population and that's what we are going to learn next. So, let's start with the ideal but the real scenario.

This is where one sample Z-test comes in. The single condition for our Z-test is that true standard deviation of the population which is sigma, it has to be known. This is an ideal situation.

Generally, population standard deviation is not known to us. When you say population standard deviation is known, it means that population mean is also known. When you know population mean, then there is no point of doing the hypothesis testing right because you are doing the hypothesis testing to test for the mean itself.

So, generally sigma is unknown but this is an ideal situation wherein sigma is known and we use one sample Z-test and in that case, the Z-test formula is how much your sample statistic is from your population statistics divided by standard error. This is a general formulation. So, at top, what is the deviation between your sample statistic to population statistics? In our case, it is sample mean to population mean divided by the standard error which is sigma by root n. So, this formulation is used only when we are using Z-test and we use Z-test only when sigma is known.

So, that's the core of it. We use specifically Z-test when sigma or the standard deviation of the population is known to us. Now, scenario 2 is when sigma is unknown.

This is more practical scenario because in practical scenarios, you will not be knowing about your population standard deviation. So, in that case, what we do is we estimate population standard deviation using sample. That's what is represented as S in your formula at the bottom where I am highlighting.

If you see, the numerator still is the same, the deviation of sample statistics from population statistic, x-bar-mu, so mu0 divided by S by root n which is the standard error. So, in the cases wherein sigma is unknown, we will directly use Z-test. Now, this is a simple flowchart just to help you understand when you are answering a question about your population.

In the cases wherein the question is restricted to means of the population, in that case, if the standard deviation is known, then you will use Z-test. If the standard deviation is unknown, you use Z-test. It's a simple classification, right? Now, this is the explorer here and what we are going to do is we can actually change the sample means if you see here and you see your Z-test value and t-test value, how does it change in sample mean and we can change sample deviation and you can see that how does the test statistics, Z-test statistics, they change using the sample deviation and sample mean.

So, in this simple case, if you think of, we have Z-distribution which is green and we have the red distribution, right, which is our t-distribution. Now, in the cases wherein the population standard deviation is known, okay, that's the green distribution is what we follow and when the population distribution is unknown, this red distribution is what we follow. Now, you could say that they are very, very identical, right, and actually that's how similar these two distributions are.

When you will have more and more samples and you actually would have more sample sizes, not more samples, when you have more sample sizes, you would see that your t-statistics or t-distribution, it actually becomes very similar to what your Z-distribution is, okay. They right now are also looking very identical, okay. Now, let's say for example if we change sample mean, you see that your Z-statistic changes because your formula right now is calculating the deviation between X bar and what your population average is, when whatever you have coded in your alternate null hypothesis.

In your null hypothesis, you have mu is equal to 50, you are taking a difference between 48 minus 50 divided by standard error, that's where you get your Z value and then you calculate the probability that Z is greater than alpha, which is where you get the p-value as 0.4, okay. Now, on the right-hand side, you see the t-test value. Now, with changes in the sample mean, you see that if sample mean is far away from the population mean, in those cases, obviously your p-values of both the tests are going to be lesser than 0.5, which means that alternate hypothesis is true, which means that the average is not equal to 50, it is lesser than 50, right.

In the right-hand side cases also, it would happen, let's say for example if the average is more than 50, then in that case also you see that p-values are actually lesser than 0.05 in both the cases, right. So both the tests are really good, they would detect the testing procedure that you are trying to do, but the only differentiating factor between two is when to use which, when your population standard deviation is known, go for Z-test, when your population standard deviation is not known, go for t-test, that's all, right. Now, this is a question, you have a pizza delivery time, a pizza shop claims that the average delivery time is 30 minutes or less, we suspect that it's actually more, okay.

Now whatever your suspicion is, you will always have that in your alternate hypothesis, that's what I told you in previous videos, if you remember, right. Your average is greater than 30 and in the first case, whatever you do not have in your alternate hypothesis, then you will frame that as your null hypothesis. So null hypothesis is less than or equal to 30, okay.

Generally, your null hypothesis would always have somewhat equals to form, greater than equals to, less than equals to or equals to, all of them would always be your null hypothesis form, okay. Now, once you have framed your hypothesis, the step two is to set a significance level because it has been not given in the question, we will assume a significance level of 5%, that's why alpha is equal to 0.05 and then we do the statistic calculation. Remember in this question, standard deviation is not given, right.

Because standard deviation is not given, hence we are going to use t-test. We have t-test, which is nothing but x bar minus mu naught, we have already been given the average time, 30, mu naught value is 30 and we take the average of all the samples which might have come out to be 33.5, which we'll feed in as x bar, 33.5 minus 30 divided by sample standard deviation, which is 6, that is, let's say, is already given or we have calculated from the samples, okay, n2 divided by sigma by root n, which is s by root n, now you will feed in the value of n here as 16 and you get this formulation out and the values look like 2.33. Now, given that t statistic value is 2.33, we can use either directly 2.33 value from the t-table, so you have t is equal to 2.33 with, we can refer to a one-tailed t-test and see for the value of t in that t-table across 5% level of significance and degrees of freedom n minus 1, which would be 16 minus 1, which is 15 and we can see if t is greater than or t is lesser than that table value, what is that range, right. Otherwise, what we can also do is we can calculate the p-value and in this case, calculated p-value comes out to be 0.17, okay, because calculated p-value is less than 0.05, hence your alternate hypothesis is true, okay, so this is what eventually happens.

As I told you, this entire hypothesis testing framework is a four-step process, first you will lay down your hypothesis, second you will specifically write what is your level of significance, if it is not given, you will assume that, then you will collect your data or create your sample statistics value, find out sample standard deviation from the data and then you will calculate your t or z statistics, if sample standard deviation is known, you will calculate z, otherwise you will calculate t. The formula still looks very same, it's just that instead of standard deviation, we use population standard deviation, we use sample standard deviation here, right. Once we have found the t-value, we have two approaches, either you can go ahead and find whether the table value of t statistics is closer to what you have, if your calculated value is greater than the value that you have in the table, then you will reject null hypothesis and you will say that you accept alternate, okay, otherwise what you can do is you can calculate the p-value which might come in, if that p-value comes out to be lesser than 0.5, 0.05, then you will say that your alternate hypothesis is eventually true. So this is the entire approach, I hope it is clear with everyone and just to summarise everything we have learnt, we started by understanding which scenario, if you are testing sample means, then you will go for z or t-test basis, if you know about population standard deviation, if you know about population standard deviation, you will go for z-test, if you don't, you go for t-test, right.

The one critical question that you ask is, is the population standard deviation known? This is the only defining criteria to test between z-test and t-test, that's all we learnt in this session, I hope this was useful to everyone. In summary, one sample test allows us to evaluate population claims using real data. By applying z-test or t-test appropriately, we can reach reliable conclusions about whether the evidence supports or contradicts the null hypothesis.

This completes our study of high-precise testing for a single mean.

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